Uniformly Accelerated Motion

Kinematics of straight-line motion with constant acceleration · problem set 1 · 15 July 2026
Dad's copy: tags, answers, references

The rules of the game

For **every** problem, including the warm-up:

1. Choose the positive direction and set $x_0 = 0$ at the start.
2. Solve for what is asked.
3. Draw the three graphs against time: $a(t)$, $v(t)$, $x(t)$ — axes labelled, key values marked (initial/final speeds, times, distances).

Graphs need not be to exact scale, but the *shapes* must be right: with constant acceleration $a(t) = a_0$ is a horizontal line, $v(t)$ is a straight line with slope a_0, and $x(t)$ is a parabola (bending up if $a_0 > 0$, down if $a_0 < 0$). Use $g = 10$ m/s^2.

The basis — constant acceleration and the resulting velocity and position:

$$\text{(i)} \quad a = a_0 = \text{const}$$

$$\text{(ii)} \quad v = v_0 + a_0 t$$

$$\text{(iii)} \quad x = x_0 + v_0 t + \tfrac{1}{2} a_0 t^2$$

x_0 and v_0 are the position and velocity at $t = 0$; a_0 is the constant acceleration; $x = x(t)$ and $v = v(t)$. Choose the positive direction first, so the signs of v_0 and a_0 follow from that choice. **Only in this copy:** the [given → find] tags on the problems — working that out himself is part of the training.

Problems

0. Warm-up (done in class — now add the graphs). A train enters a station at speed v_0. The station has length L, and the train stops exactly at its far end, decelerating uniformly. Treating the train as a point, find the acceleration and the stopping time.

[given v_0, x=L, v=0 → a_0, t]

1. Away from the lights. A car starts from rest and accelerates uniformly at 2 m/s^2 for 6 s. Find its final speed and the distance covered.

[given v_0=0, a_0, t → v, x]

2. Between stations. A train speeds up from 10 m/s to 25 m/s in 30 s at constant acceleration. Find the acceleration and the distance travelled.

[given v_0, v, t → a_0, x]

3. The cyclist. A cyclist moving at 6 m/s accelerates uniformly at 2 m/s^2 along a 16 m stretch of road. Find his speed at the end of the stretch and the time it took.

[given v_0, a_0, x → v, t]

4. Braking distance. A car travels at 20 m/s. The driver brakes, and the car decelerates uniformly at 5 m/s^2. (a) How long and how far does it travel before stopping? (b) If the initial speed were doubled (40 m/s), how far would it travel? How many times longer is that?

[given v_0, v=0, a_0 → t, x]

5. **Off the bridge.** A stone is released from rest from a bridge and hits the water after 2 s. How high is the bridge, and how fast does the stone hit the water? (Ignore air resistance.)

[given $v_0{=}0$, $a_0{=}g$, $t \rightarrow v, x$]

6. **Straight up.** A ball is thrown straight up at 15 m/s. (a) How long does it take to reach the top? (b) What is the maximum height? (c) When does it return to the hand, and with what velocity? — On your graphs, pay special attention: what is the acceleration *at the very top*?

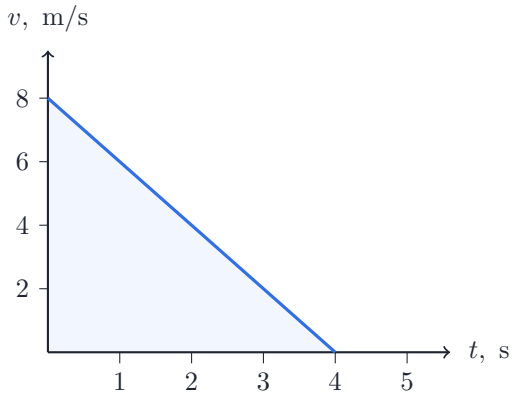
[given v_0, $a_0{=}-g \rightarrow t, x, v$]

7. **The whole trip (graphs first!).** A bus leaves a stop: it accelerates uniformly from rest to 12 m/s in 4 s, drives at a steady 12 m/s for 6 s, then brakes uniformly to a stop in 3 s. Draw all three graphs *first*; then find the acceleration and the distance for each phase, the total distance, and the average speed of the trip.

[three phases $\rightarrow$ piecewise graphs]

8. **Read the graph.** The v–t graph of a tram is shown. From the graph, find the initial speed, the acceleration (slope), and the distance travelled (area under the graph). Then sketch the matching $a(t)$ and $x(t)$.

[given graph $\rightarrow v_0, a_0, x$]



9. **Challenge: a useful consequence.** Using a v–t graph for uniform acceleration, show that the displacement equals the area of a trapezium: $x - x_0 = \frac{1}{2}(v_0 + v)\,t$. Then use this to re-derive your answer to Problem 0 in one line.

[derivation]

Answers

0. $a_0 = -\dfrac{v_0^2}{2L}$; $\quad t = \dfrac{2L}{v_0}$.

1. $v = 12$ m/s; $\quad x = 36$ m.

2. $a_0 = \dfrac{25 - 10}{30} = 0.5$ m/s^2; $\quad x = 10 \cdot 30 + \frac{1}{2} \cdot 0.5 \cdot 30^2 = 525$ m.

3. $16 = 6t + t^2 \Rightarrow t = 2$ s; $\quad v = 6 + 2 \cdot 2 = 10$ m/s.

4. (a) $t = 4$ s, $x = 40$ m; (b) $t = 8$ s, $x = 160$ m — 4× the braking distance.

5. $x = \frac{1}{2} \cdot 10 \cdot 2^2 = 20$ m; $\quad v = 20$ m/s.

6. (a) 1.5 s; (b) 11.25 m; (c) returns at 3 s with 15 m/s downward. At the top $v = 0$ but $a_0 = -g = -10$ m/s^2 — the $a(t)$ graph is one unbroken horizontal line.

7. Phase 1: $a_0 = +3$ m/s^2, $\Delta x = 24$ m; phase 2: $a_0 = 0$, $\Delta x = 72$ m; phase 3: $a_0 = -4$ m/s^2, $\Delta x = 18$ m. Total 114 m in 13 s; average speed ≈ 8.8 m/s. The $v(t)$ graph is a trapezium; its area is the total distance.

8. $v_0 = 8$ m/s; $\quad a_0 = -2$ m/s^2; $\quad x = \frac{1}{2} \cdot 8 \cdot 4 = 16$ m (the shaded triangle). $x(t)$ is a parabola bending down, flattening out at $t = 4$ s.

9. Trapezium with parallel sides v_0 and v, width t: $x - x_0 = \frac{1}{2}(v_0 + v)t$. For the train, $L = \frac{1}{2}(v_0 + 0)t \Rightarrow t = 2L/v_0$, then $a_0 = (0 - v_0)/t = -v_0^2/(2L)$.

Graph checklist for marking: within each constant-acceleration phase, $a(t)$ is horizontal, $v(t)$ is straight, and $x(t)$ is a parabola with the right bend. At a stop or turn-around, $v = 0$ and the tangent to $x(t)$ is horizontal; no graphs jumps.

More practice from our archive

- **Siyavula, Physical Sciences Gr 10**
  `docs/textbook/siyavula-physical-sciences/Gr10_PhysicalSciences_Learner_Eng.pdf`
  ch 21 *Motion in one dimension*: worked examples throughout (PDF pp. 403–452, graph-heavy from p. 430) and *End of chapter exercises* (PDF pp. 453–457) — includes drawing and reading x–t/v–t graphs. Brief solutions in ch 25 ($\approx$ PDF p. 504).

- **NCERT, Class 9 Science**
  `docs/textbook/ncert-class9-physics/ch04-describing-motion.pdf`
  §4.2 graphs of motion (PDF pp. 9–16), §4.3 kinematic equations (PDF pp. 16–18), exercises *Revise, Reflect, Refine* (PDF pp. 21–24, graph question on p. 23).

- **CK-12, Quizzes & Tests**
  `docs/textbook/ck12-physical-science/`
  `CK-12-Physical-Science-MS-Quizzes-and-Tests-with-Answers.pdf`
  quiz 12.2 *Speed and Velocity* (PDF p. 161), quiz 12.3 *Acceleration* (PDF p. 164), Unit-4 *Motion and Forces* test (PDF p. 353); answers are in the same file.